2.4 Trapezium

We are familiar with parallelogram and rhombus. What will happen in a parallelogram if one pair of parallel sides are not equal? Can you draw it? How will it look like? The shape looks as given in Fig.2.19.

A parallelogram with one pair of non-parallel sides is known as a Trapezium.

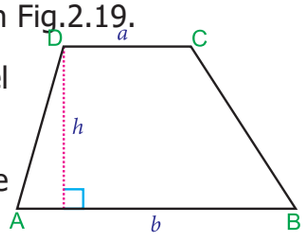


Fig.2.19

The distance between the parallel sides is the height of the trapezium.

Here the sides AD and BC are not parallel, but AB is parallel to DC.

Isosceles Trapezium

If the non - parallel sides of a trapezium are equal (AD = BC) then, it is known as an isosceles trapezium.

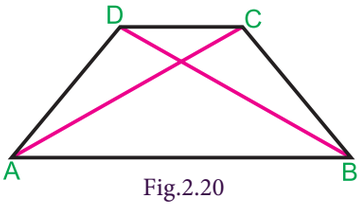


Fig.2.20

2.4.1 Area of the Trapezium

ABCD is a trapezium with parallel sides AB and DC measuring 'a'units and 'b' units respectively. Let the distance between the two parallel sides be 'h' units. The diagonal BD divides the trapezium into two triangles ABD and BCD.

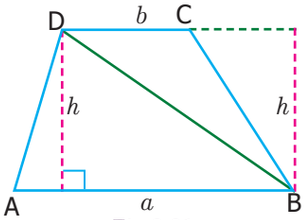


Fig.2.21

Area of the trapezium = area $\triangle$ of ABD + area of $\triangle$ BCD

$= \frac{1}{2}$ x AB x h + $\frac{1}{2}$ x DC x h [since the two triangles ABD and BCD have same heights]

Area of a trapezium $= \frac{1}{2}$ x $\{h$ x (AB + DC)$\}$

Therefore, Area of the trapezium = $\frac{1}{2}$ x h $(a+b)$ $sq.units.$

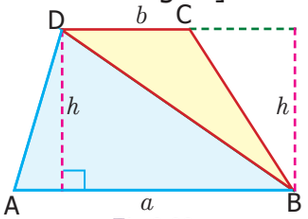


Fig.2.22

Example 2.10

Find the area of the trapezium whose height is 14 cm and the parallel sides are 18 cm and 9 cm of length.

Solution

Given, height (h) = 14 cm

parallel sides are (a) = 18 cm and (b) = 9 cm

Area of the trapezium $= \frac{1}{2}$ x h $(a+b)$

$= \frac{1}{2}$ x 14 (18+9)

$= 7(27)$

$= 189$ $sq.$ cm

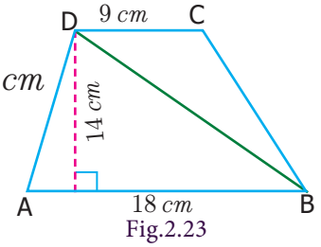


Fig.2.23

Therefore, area of the trapezium is 189 $sq.$ $cm.$

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